

Exercise 03

Written Part1

1. The following binary numbers presents positive integers(i64)

- 1011001_2
- 0110011_2
- 101001000010001_2
- 110111100110011_2

(a) Convert the binary numberse to their decimal values (show your work)

- $1011001_2 = 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^2 + 1 = 64 + 0 + 16 + 8 + 0 + 0 + 0 + 1 = 89$
- $0110011_2 = 0 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 = 0 + 32 + 16 + 0 + 0 + 2 + 1 = 51$
- $101001000010001_2 = 1 \cdot 2^{14} + 1 \cdot 12 + 1 \cdot 9 + 1 \cdot 2^0 = 16384 + 4096 + 512 + 1 = 21009$
- $110111100110011_2 = 1 \cdot 2^{14} + 1 \cdot 2^{13} + 1 \cdot 2^{11} + 1 \cdot 2^{10} + 1 \cdot 2^9 + 1 \cdot 2^8 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^1 + 1 \cdot 2^0 = 28467$

answer : 89, 51, 21009, 28457

(b) Find the two's complement representation of those binary numbers

2's complement :

- 1011001_2

Step	Operation	Binary Result
One's Complement	Invert all bits	010 0110
Two's Complement	Add 1	010 0111

- 0110011_2

Step	Operation	Binary Result
One's Complement	Invert all bits	100 1100

Two's Complement	Add 1	100 1101
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- 101001000010001_2

Step	Operation	Binary Result
One's Complement	Invert all bits	010 1101 1110 1110
Two's Complement	Add 1	010 1101 1110 1111

- 110111100110011_2

Step	Operation	Binary Result
One's Complement	Invert all bits	001 0000 1100 1100
Two's Complement	Add 1	001 0000 1100 1101

2. Patterns

(a) How many different bit patterns can be represented using 63 bits?

| 2^{63}
which is approximately 9,223,372,036,854,775,808

(b) If we want to store any integer x where $0 \leq x \leq 25$. What is the smallest number of bits we can use?

$$2^n \geq 26 \rightarrow n = 5$$

| Thus, we need 5 bits

(c) If we represent colours as red-green-blue (RGB) triples, where we use 4 bits for red, 4 bits for green, and 4 bits for blue. How many different colours can we represent?

| If we used 4 bits each for each channel : $2^4 = 16$ values.
Thus, for RGB, $\frac{\text{bits}}{\text{digit}} \times 4 \text{ digit} = 16$ bits combinations of colors can be represented .

(d) If we represent a decimal number (like 13579) by concatenating the binary representations of each decimal digit. How many bits would we need for a number which contains 4 digit

Digits : 0~9

$9 = 1001_2 \Rightarrow 4$ bits. So each digit needs 4 bits.

Number that contains 4 digits will need $4 \frac{\text{bits}}{\text{digit}} \times 4 \text{digit} = 16$ bits

3. Find out which rounding mode Rust is using for floating point computations.

Rust uses round to nearest, ties to even as its default rounding mode for floating-point arithmetic (f32 and f64).

Reference : <https://doc.rust-lang.org/reference/types/numeric.html>

The types f32 and f64 correspond to the IEEE 754-2008 binary32 and binary64 floating-point formats.

Because **IEEE 754 specifies round to nearest, ties to even as the default rounding mode**, Rust arithmetic follows this behavior by default.

4. If we have a 8-bit IEEE754 type floating point number (E4M3: sign, 4-bit exponent, 3-bit mantissa) which values can we represent?

E4M3 :

Field	Bits	Notes
Sign (S)	1	0 = positive, 1 = negative
Exponent (E)	4	4-bit exponent
Mantissa / Fraction (M)	3	Fraction bits (with implicit leading 1 for normalized numbers)

reference: https://en.wikipedia.org/wiki/Extended_precision

Representable values in E4M3

- **Normalized numbers** (E = 1..14):

$$\text{value} = (-1)^{\text{sign}} \times (1 + \text{mantissa}/8) \times 2^{E-7}$$

- using above formula,

- Largest normal (E=14, M=7) : $1.875 \times 2^7 \approx 240$

- Smallest normal (E=1, M=0) : $1 \times 2^{-6} \approx 0.0156$
- **Subnormal numbers** (E = 0, mantissa $\neq 0$):

$$\text{value} = (-1)^{\text{sign}} \times (\text{mantissa}/8) \times 2^{-6}$$

- using above formula,
 - Largest subnormal (M=7) : $7/512 \approx 0.0137$
 - Smallest subnormal (M=1) : $1/512 \approx 0.00195$
- **Zero**: E = 0, mantissa = 0 $\rightarrow \pm 0$
- **Infinity**: E = 15, mantissa = 0 $\rightarrow \pm \infty$
- **NaN**: E = 15, mantissa $\neq 0 \rightarrow \text{NaN}$